Quiz 1, Modules 1–2 (paper)

Module 1 - Introduction & Math Review · Canvas quiz · 26 Aug 2026

Name _____ Date _____

Question 1 Which of the following is true about a process in which the rate depends on the concentration? Select all that apply.

A. The half-life depends on the concentration of the drug present.
B. The half-life is independent of concentration.
C. The rate equation is $dC/dt = -kC$.
D. The rate equation is $dC/dt = -k$.

Question 2 Which of the following is true about a zero-order process?

A. The elimination rate increases as a function of time.
B. The elimination rate is at its highest when the amount of drug in the body is at its lowest.
C. The elimination rate is independent of the amount of drug in the body.
D. The elimination rate is at its highest when the amount of drug in the body is at its highest.

3a A 165-lb patient received a single IV bolus dose of an antibacterial agent at a level of 30 mg/kg. Blood samples were collected at various time intervals. The elimination of drug seems to follow a first-order model best described by the equation $C = C_0 e^{-kt}$. The plasma concentrations (C_p, mcg/mL) were: 0.5 hr, 112; 2 hr, 90.9; 4 hr, 68.9; 8 hr, 39.6; 16 hr, 13.1. What is the half-life of this agent in this patient?

3b A 165-lb patient received a single IV bolus dose of an antibacterial agent at a level of 30 mg/kg. The elimination of drug follows $C = C_0 e^{-kt}$, and the plasma concentrations (C_p, mcg/mL) were: 0.5 hr, 112; 2 hr, 90.9; 4 hr, 68.9; 8 hr, 39.6; 16 hr, 13.1. What is the volume of distribution of this agent in this patient?

3c A 165-lb patient received a single IV bolus dose of an antibacterial agent at a level of 30 mg/kg. The elimination of drug follows $C = C_0e^{-kt}$, and the plasma concentrations (C_p, mcg/mL) were: 0.5 hr, 112; 2 hr, 90.9; 4 hr, 68.9; 8 hr, 39.6; 16 hr, 13.1. How much time following administration of the dose is required for 75% of the dose to be eliminated?